WORKSHEET · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 8: Models and Agents

Module 3 | 45 minutes | Modelling | Grades 6–8 · ages 11–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. model ___ A. Patterns that arise from many simple rules.
b. time step ___ B. An individual following rules (often a clone).
c. agent ___ C. One update of the model (e.g. a year).
d. emergence ___ D. A simplified version of reality used to predict.

2. Plan your code

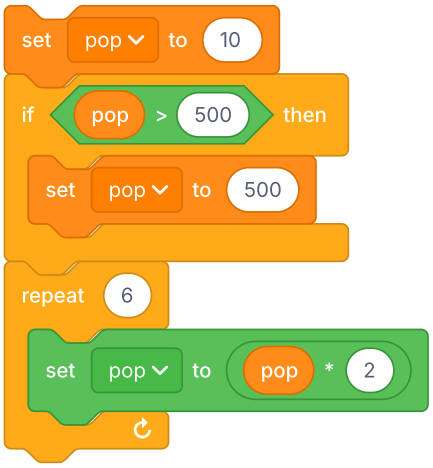
Before you build it: Rabbit model: set pop to 10. Repeat 6 years: set pop to pop × 2. But food is limited: if pop > 500, set pop to 500. Say the final population (500). Write or draw the blocks you will use.

3. Plan your code

Before you build it: Particles in a box: the Particle hides and makes 15 clones. Each clone goes to a random position, points in a random direction, shows, and moves 5 steps forever, bouncing off edges. Write or draw the blocks you will use.

4. Debug the model

The rabbit model's population keeps going past 500. What is wrong? Circle one.



a. A. The limit check is outside the loop, so it only runs once at the start
b. B. It should be > 640
c. C. pop should start at 0

5. Explain your thinking

Every model leaves things out. Name two things the rabbit model ignores, and explain how you would add one of them as a new rule.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can build a step-by-step model.
☐ I can build an agent-based model with clones.
☐ I can explain what a model leaves out.